

ISM working group XII

Coordinator: Sam Leitner (Maryland)

Working group participants:

Code	Cosmological zoom-in simulator	Code contacts in group XII
ART-I	Daniel Ceverino	<u>Daniel Ceverino</u>
ART-II	Sam Leitner	Nick Gnedin, Andrey Kravtsov, <u>Sam Leitner</u>
ENZO	John Wise	Nathan Goldbaum, Cameron Hummels, Ji-hoon Kim, Brian O'Shea
GADGET	Amit Kashi, Dusan Keres, Justin Read	<u>Amit Kashi</u> , <u>Dusan Keres</u> , Ken Nagamine, Brant Robertson, Emilio Romano-Diaz, Robert Thompson
GADGET-SPHS	Justin Read	<u>Justin Read</u> , Alexander Hobbs
GASOLINE	Sijing Shen	Charlotte Christensen, Piero Madau, Lucio Mayer, <u>Sijing Shen</u> , James Wadsley
RAMSES	Oscar Agertz, Romain Teyssier	<u>Oscar Agertz</u> , Matteo Tomassetti

Outline

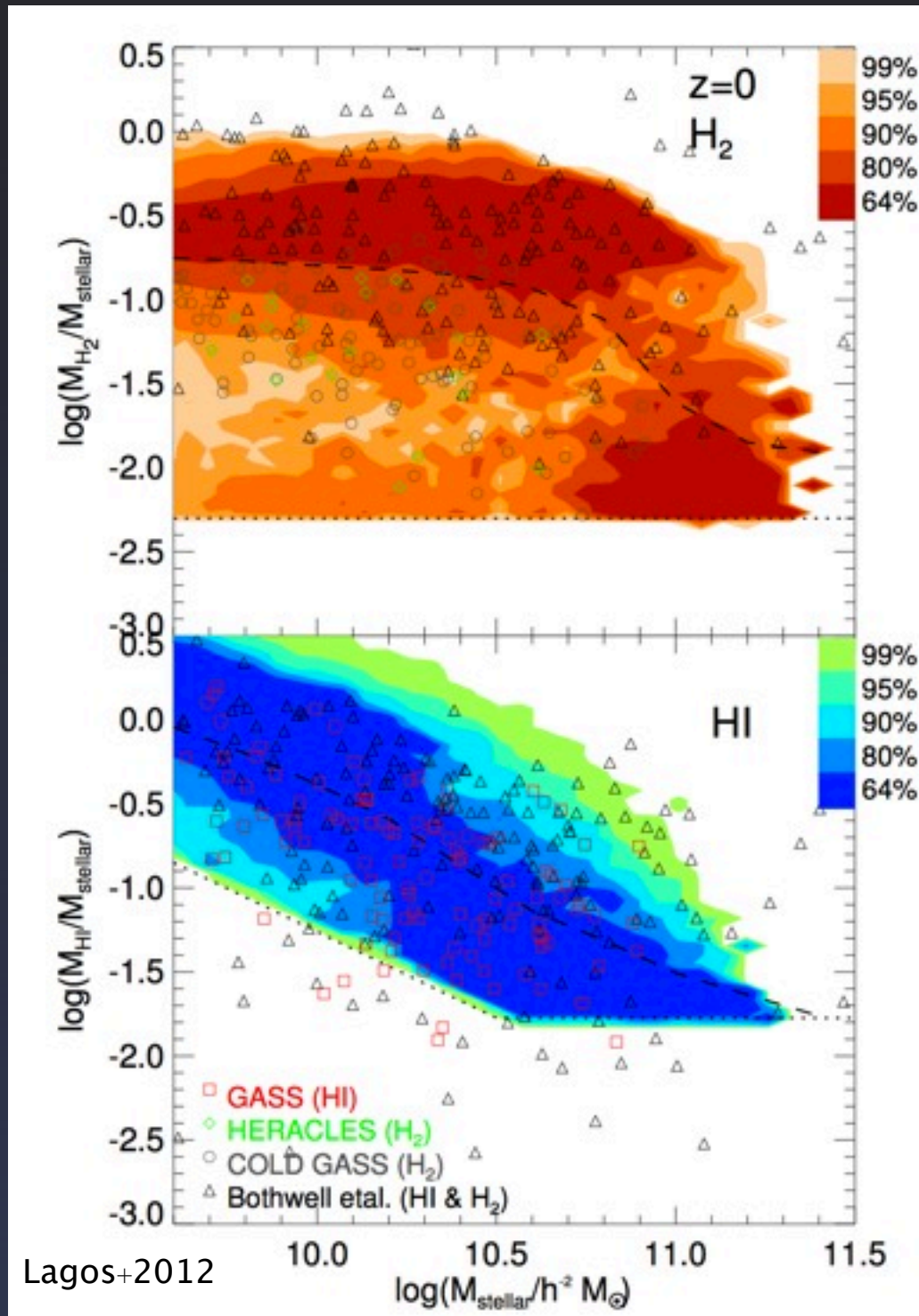
- Science objectives:
 - Interpret the simulated ISM in the context of observational constraints.
 - Understand sensitivity of predictions to method and implementation.
- Numerical objective:
 - Develop a meaningful dialog between different simulation code groups.
- Simulation plans.

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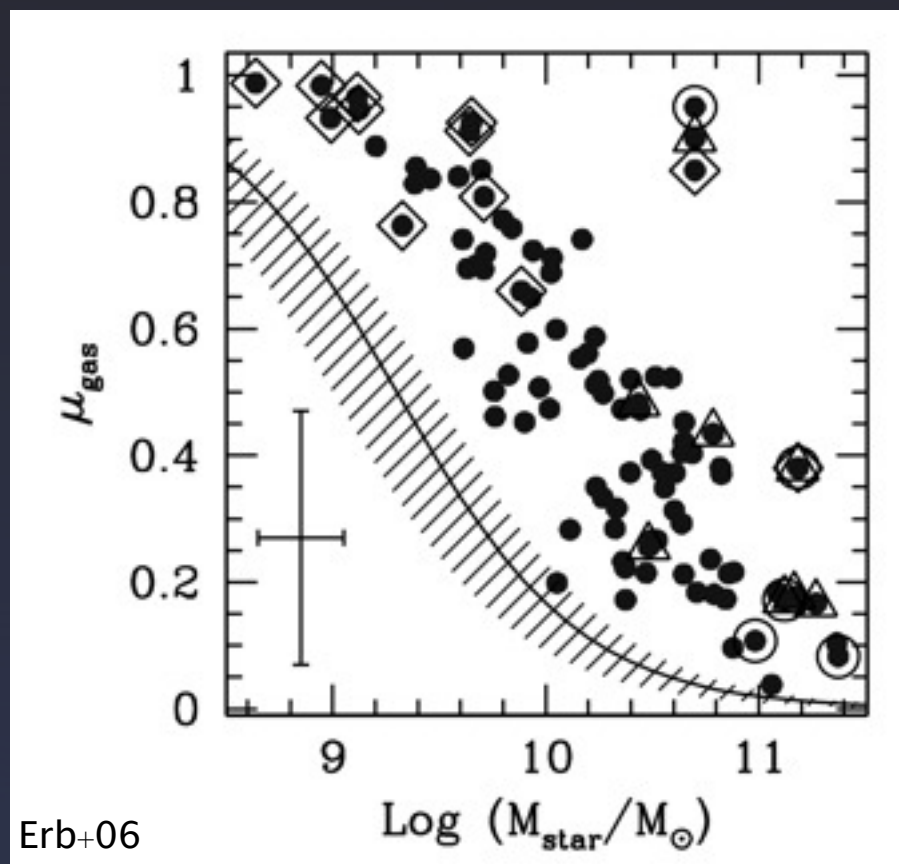
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Gas fractions (M_* , z)

HI and H₂ fractions as a function
of stellar mass at $z=0$



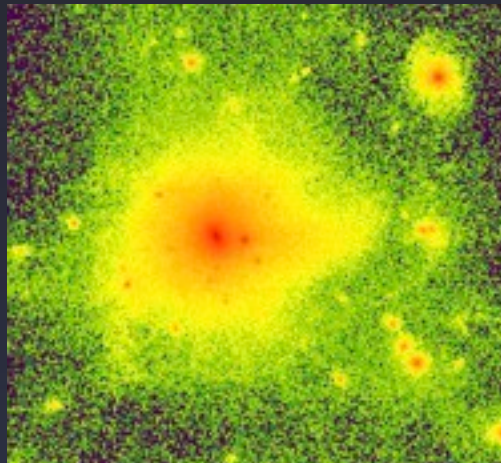
Inferred gas fraction at
higher redshift ($z=2$)



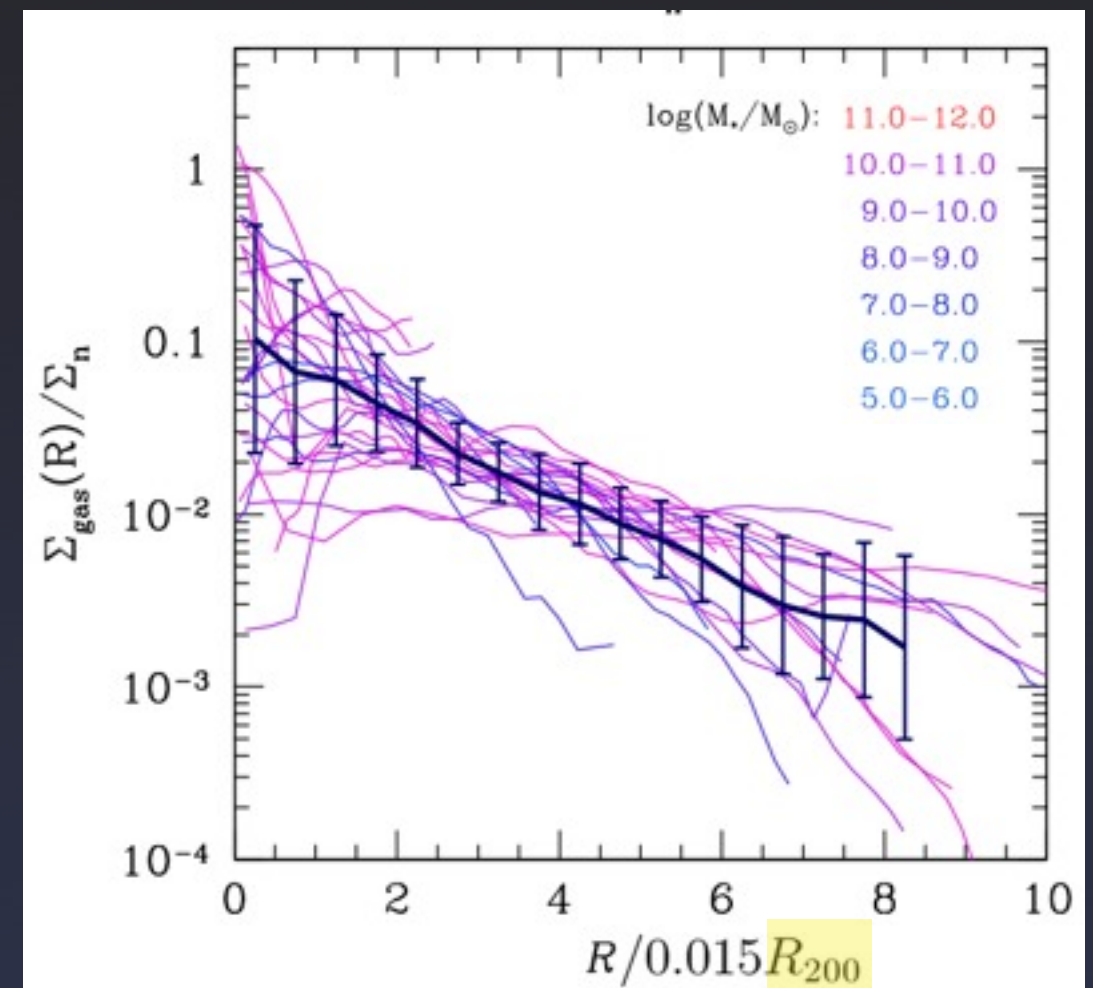
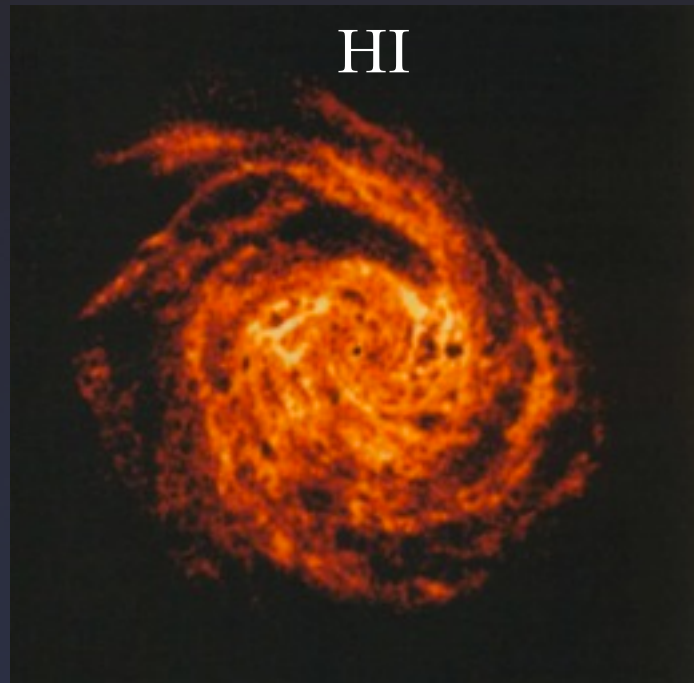
Gas disk extents

Gas scale lengths collapse when related to halo virial radius. Scatter may be tied to scatter in halo spin parameter (Kravtsov 2013).

IC



HI

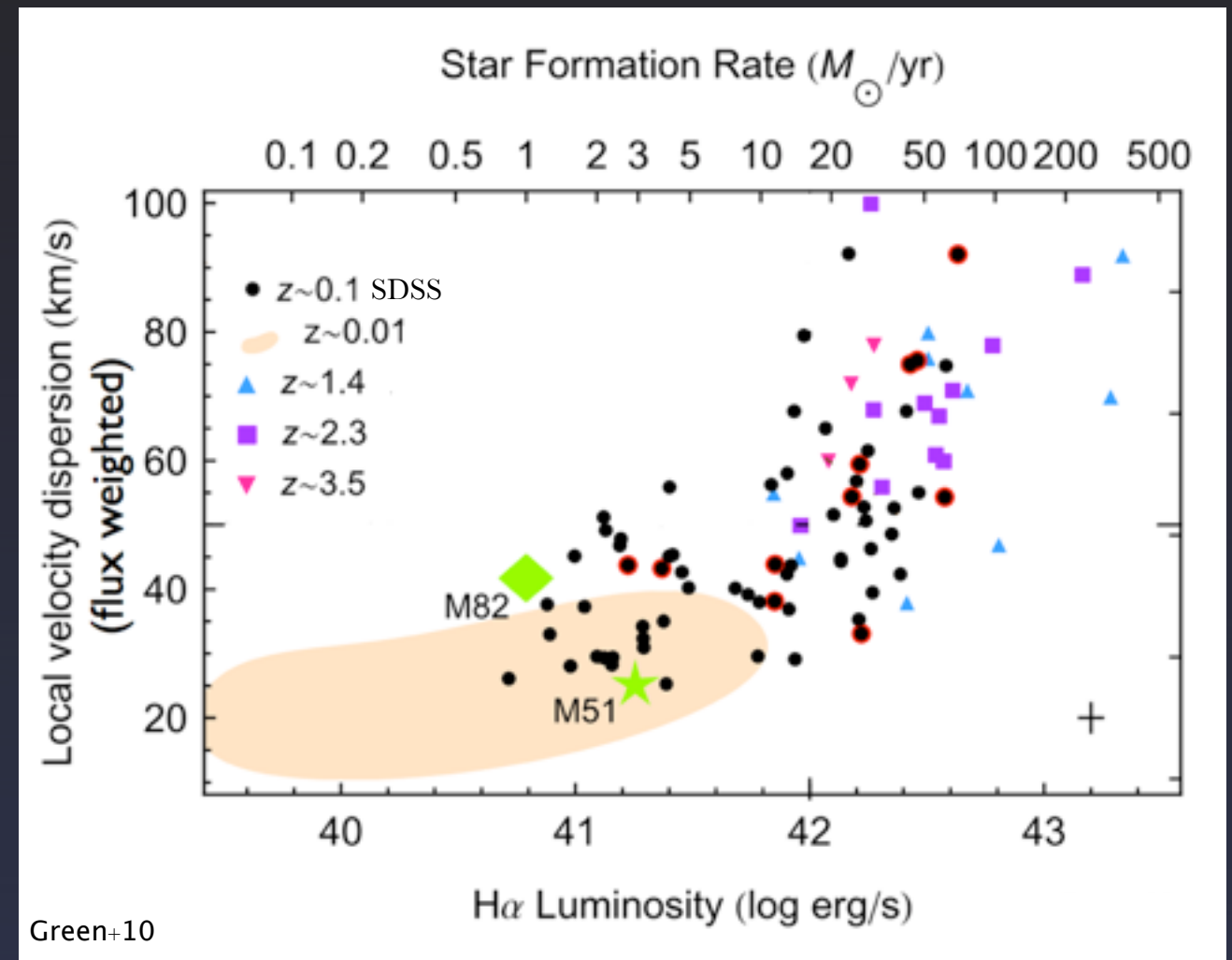
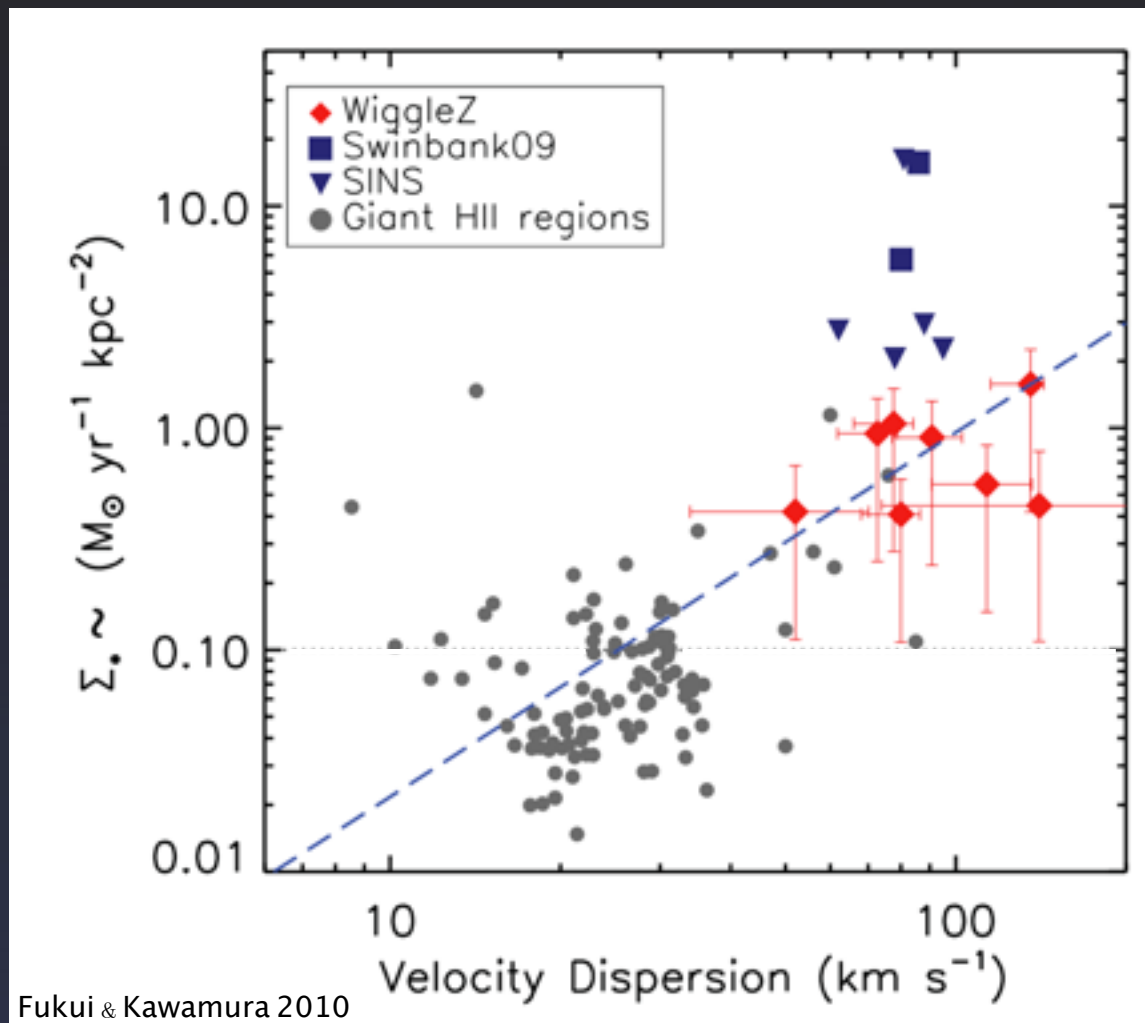


plot: Kravtsov 2013 ; data: Little THINGS, THINGS/HERACLES, Leroy+08 Zhang+12;

How robust are gas scale lengths predicted by simulation for the same ICs?

Turbulence

- Does velocity dispersion match observations of HII regions?
 - Are there spurious dependencies (redshift or stellar mass)?

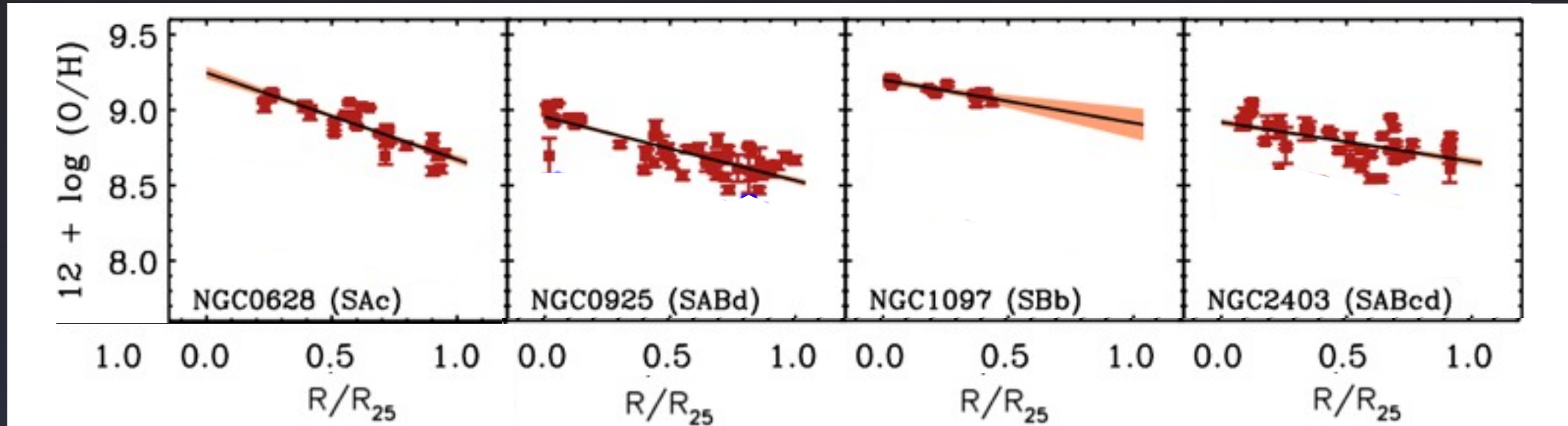


- What generates velocity dispersion in our gas disks?
- Do different feedback prescriptions self-regulate to the same gas velocity dispersion? What is the resolution dependence?

Gas-phase metallicity profile

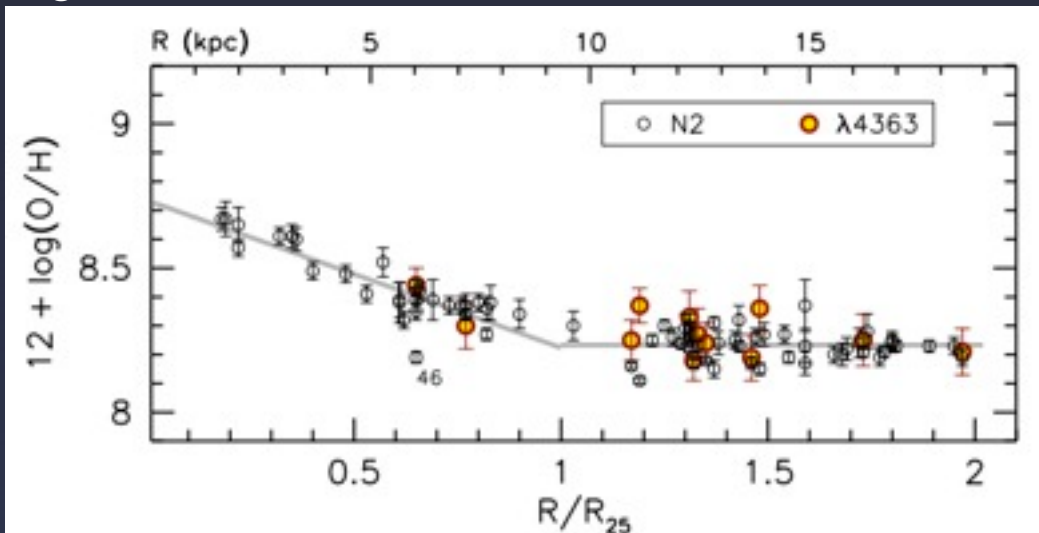
Metallicity decreases within the optical disk radius

SINGS Moustakas+10. gradient $(-0.33 \pm 0.16 \text{ dex}) / R_{25}$

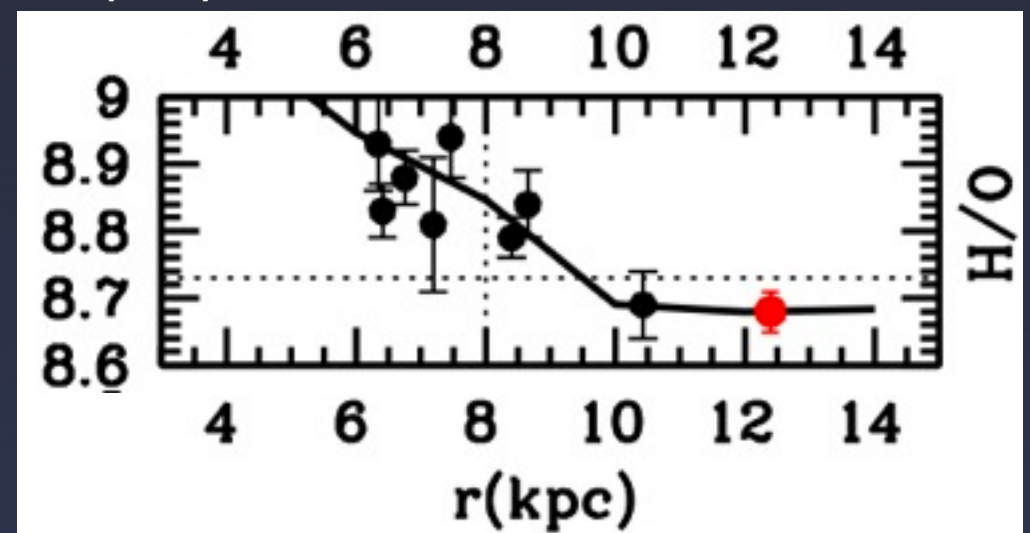


Metallicity gradient is flat in the outer gas disk

e.g., NGC 3621 Bresolin 2012



Milky Way Estaban+13

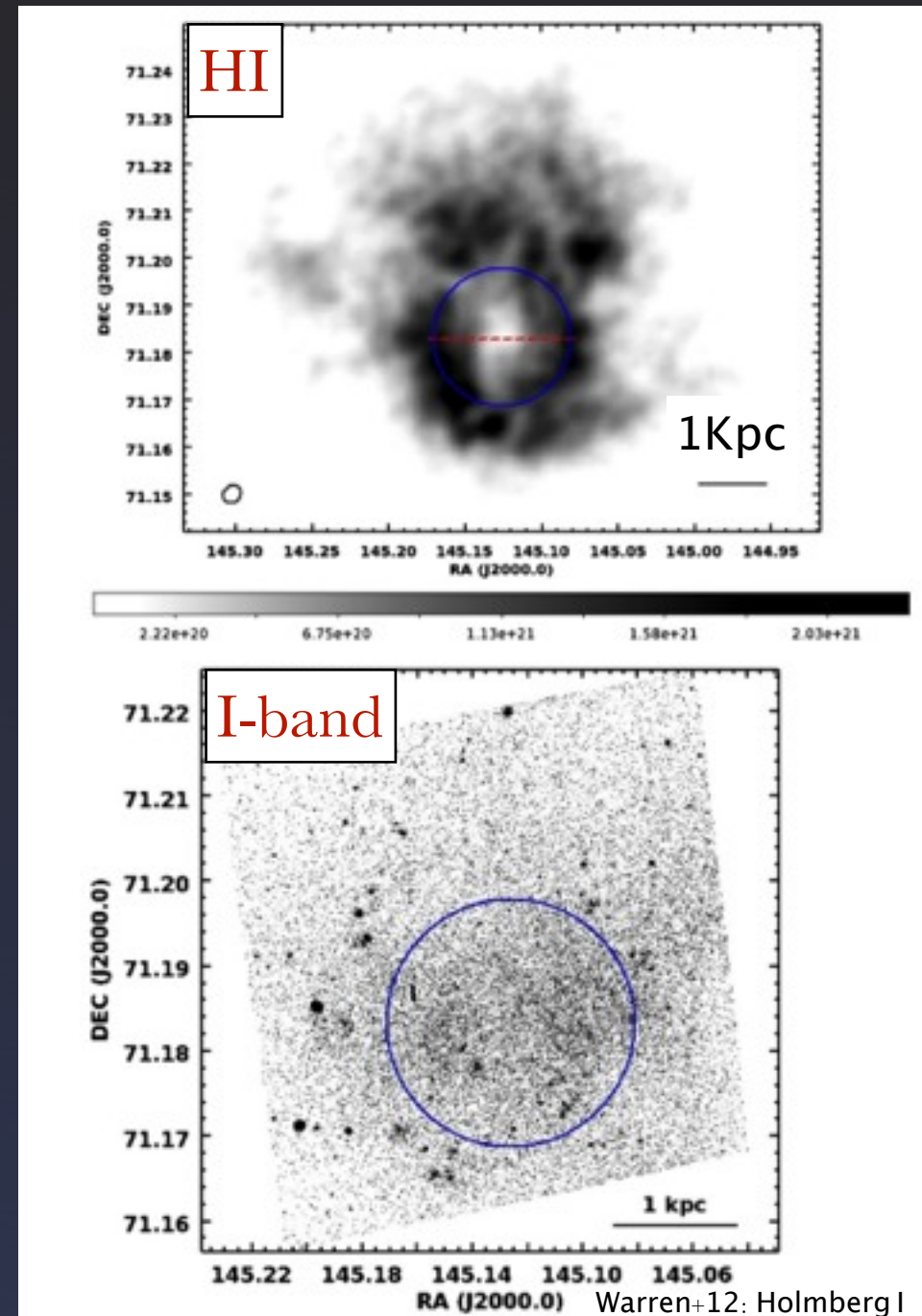
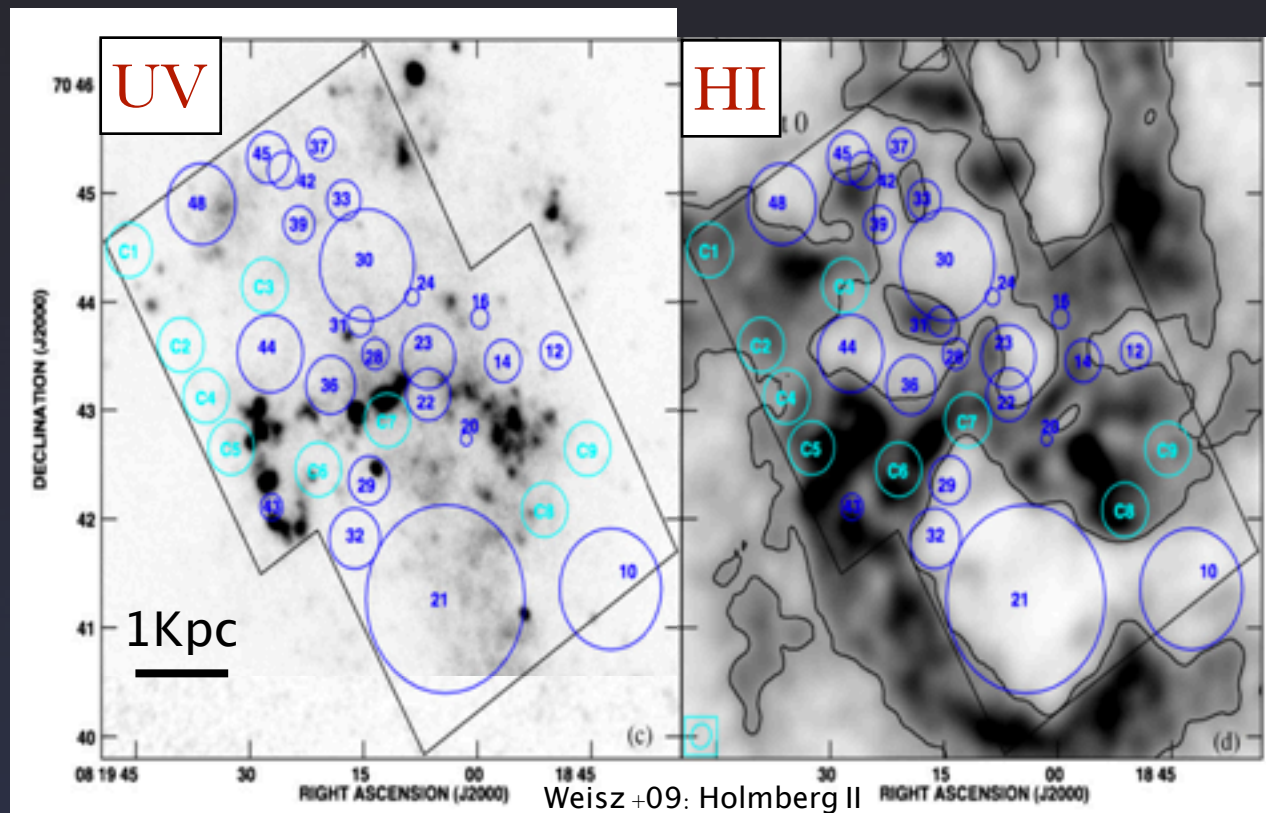


(see also Bresolin+09+12, Zahid & Bresolin 2011, Marino+12, Estaban+13)

Kiloparsec scale HI holes

Are HI holes created directly by stellar feedback (Weisz+09)?
Do they form as part of general turbulent cascade in the ISM (Hopkins 2012)?

THINGS

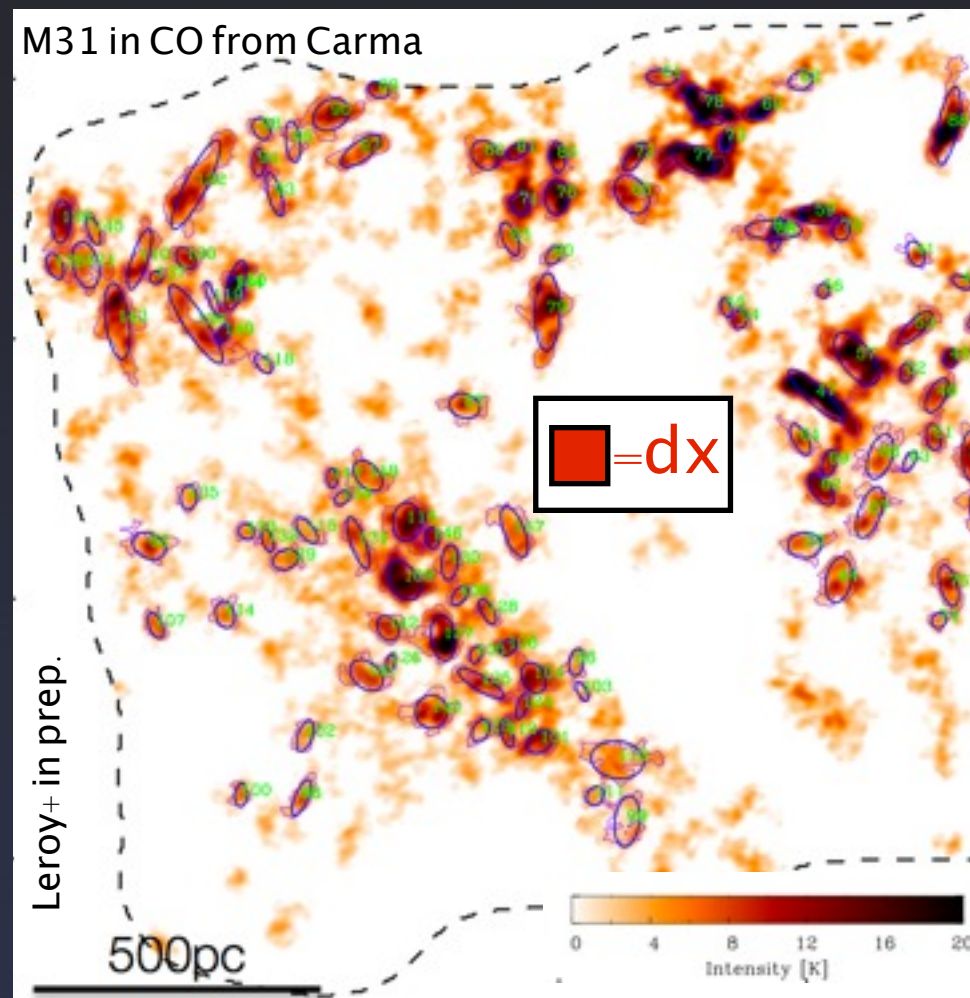


Do simulated holes result from stellar populations spread over 100 Myr or from SSPs?

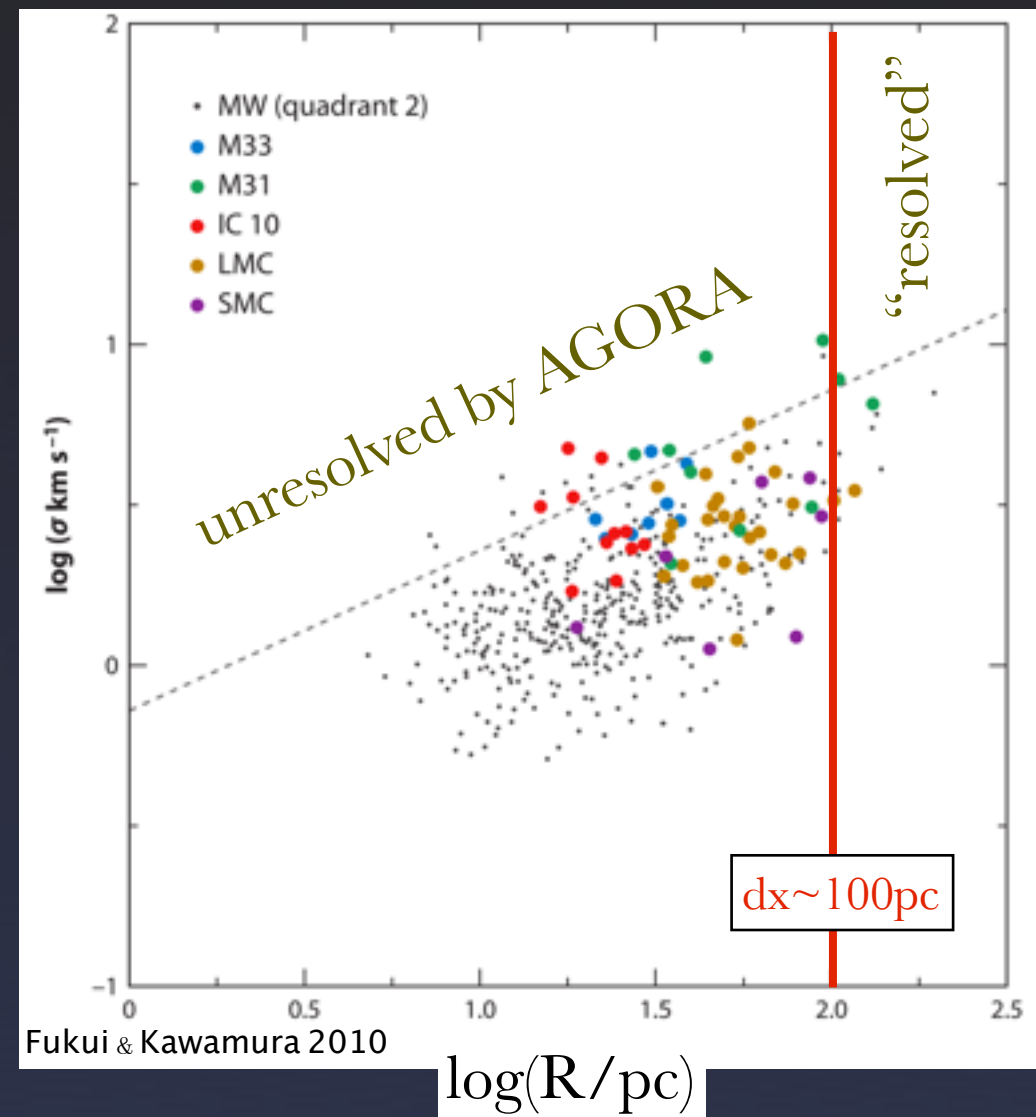
Cold clouds clumps cores

What can we NOT simulate at low z ?

Molecular clouds in spirals



Typical cloud sizes

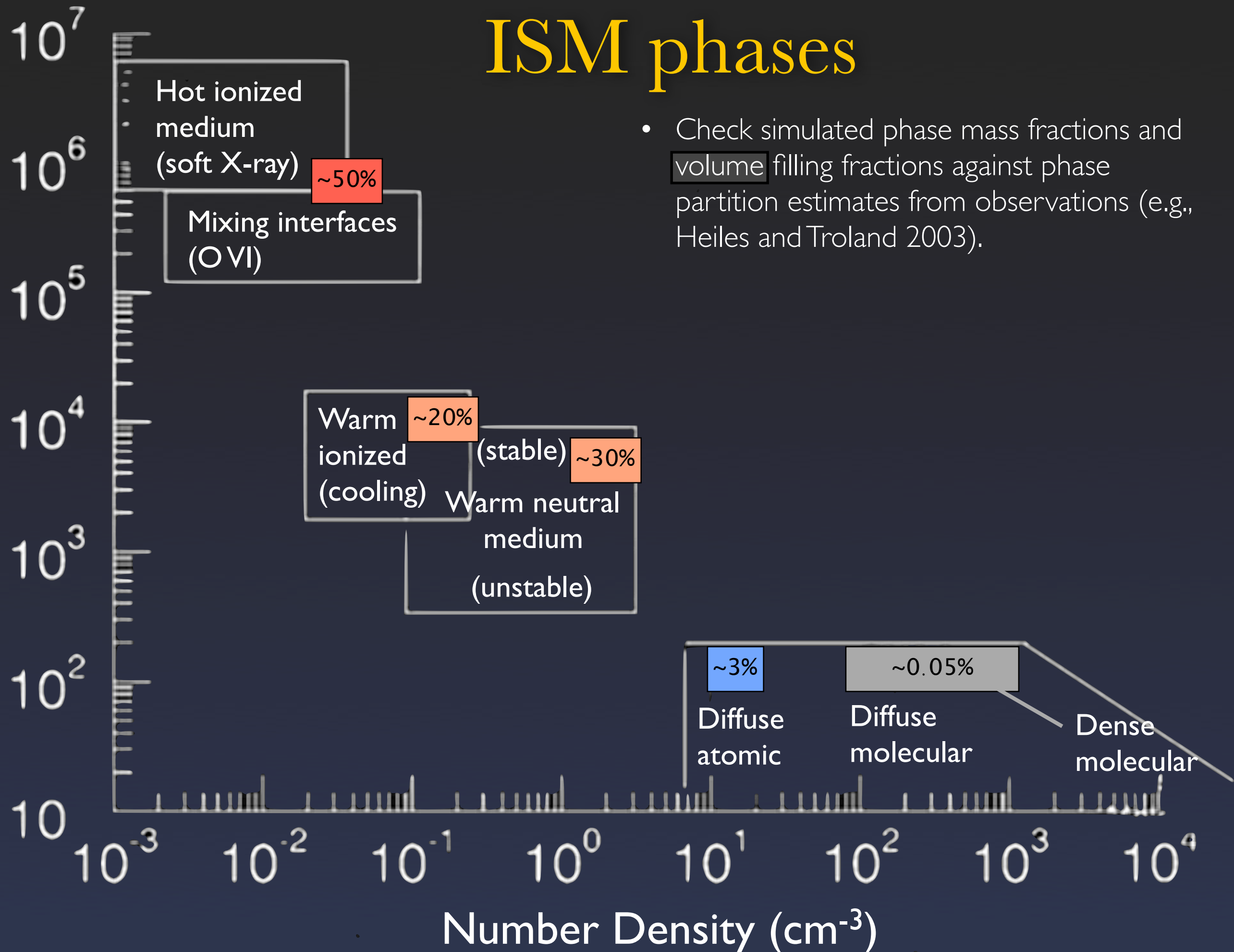


Most low redshift GMC disruption is still below our resolution limit.

ISM phases

- Check simulated phase mass fractions and **volume** filling fractions against phase partition estimates from observations (e.g., Heiles and Troland 2003).

Temperature (K)



Number Density (cm^{-3})

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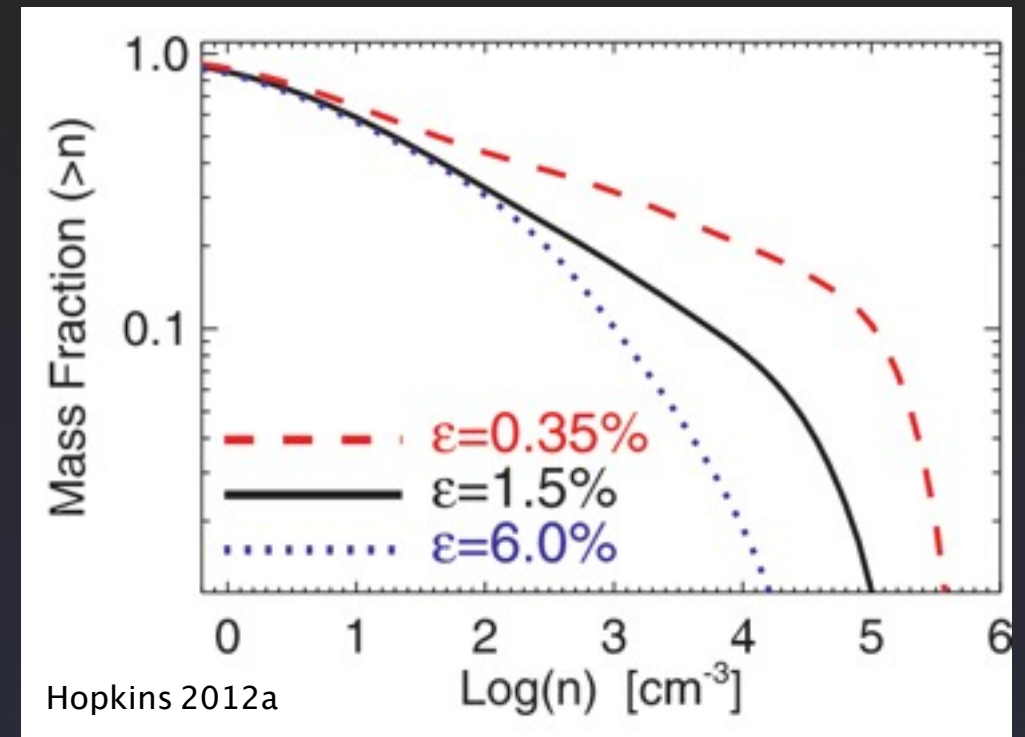
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The life of dense gas

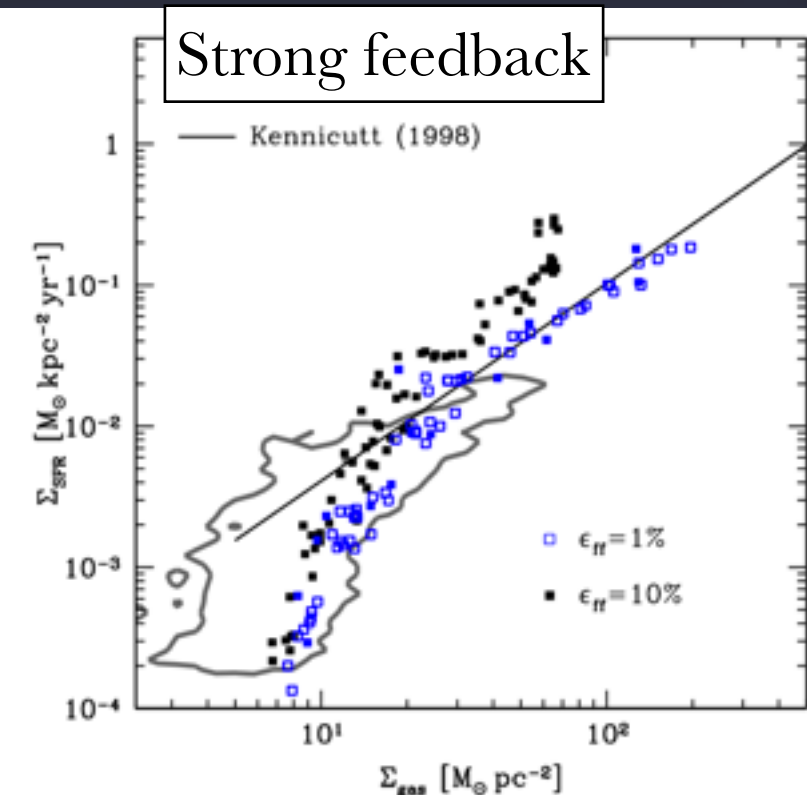
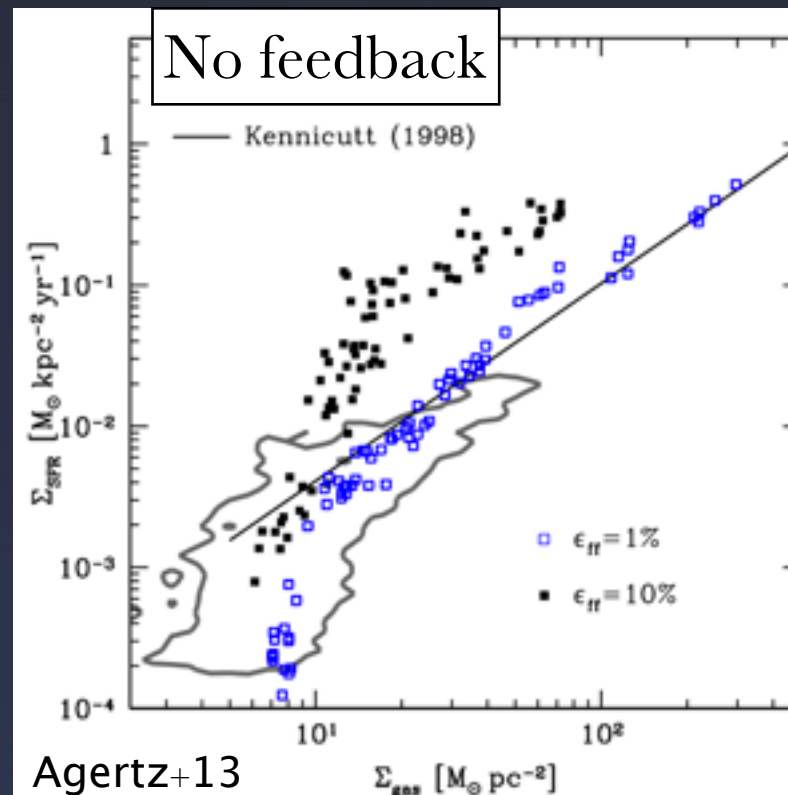
The interplay between star formation and feedback sets the dense gas distribution function (code dependence?).

$$\frac{d\rho_*}{dt} = \frac{\epsilon \rho_{\text{gas}}}{t_{\text{ff}}}$$

+ feedback



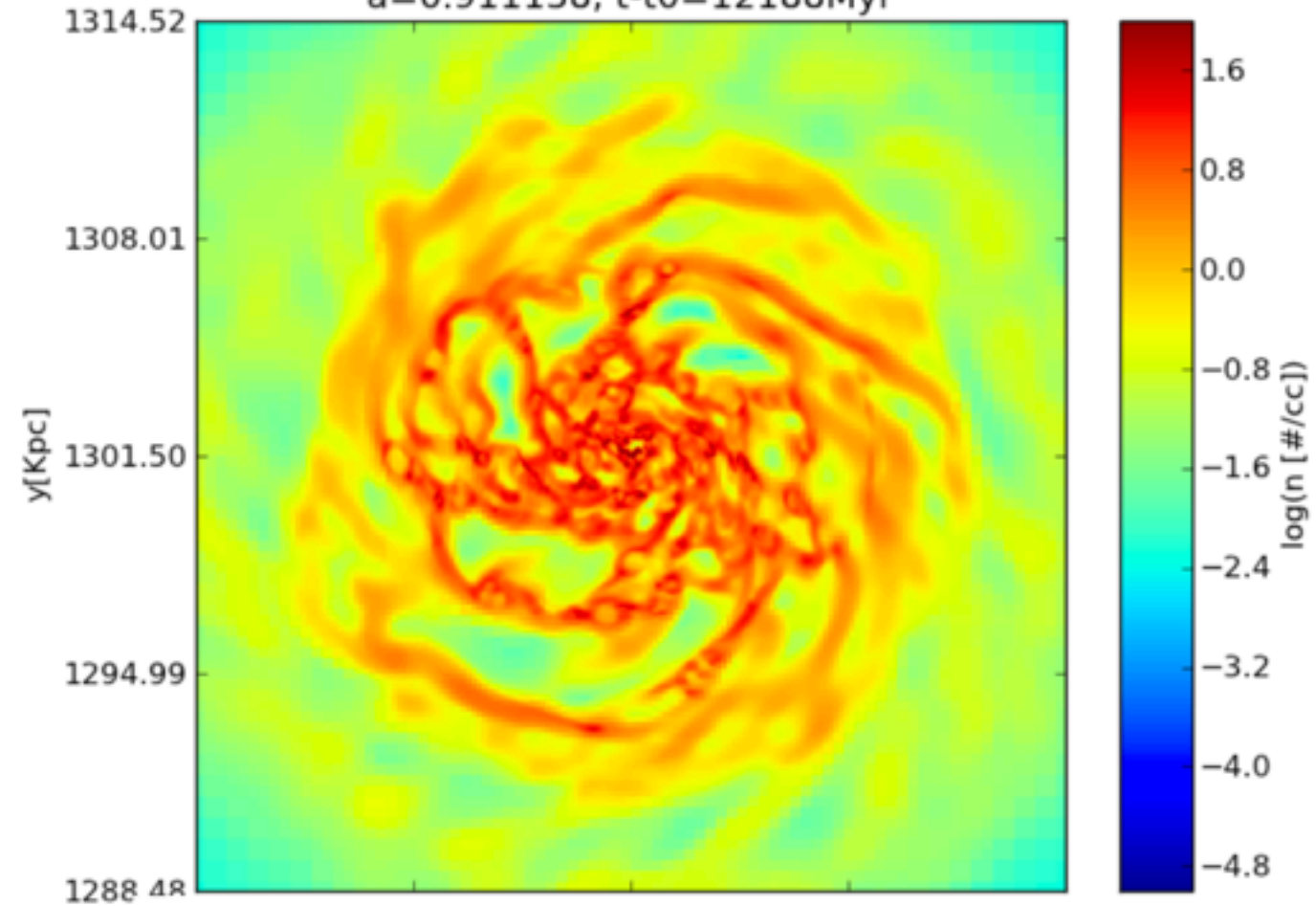
Do we see self-regulation in all “successful” simulations? How does the gas density PDF vary between codes and can we understand the variation?



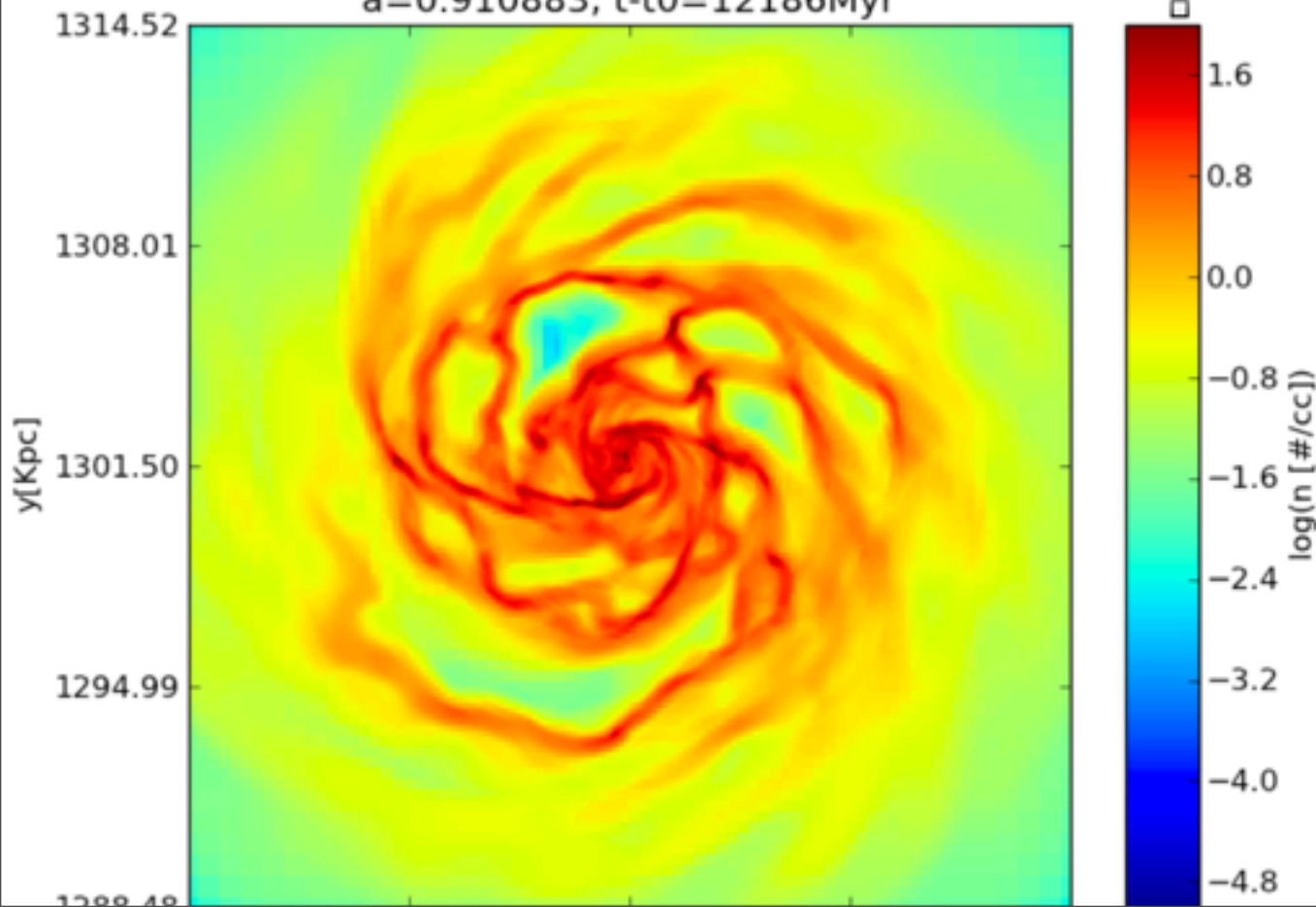
Gas disk structure

- Different star formation and feedback implementations can generate a significantly different ISM (power spectra, density PDF, etc)
- Can we share a subgrid language? Can codes with the same physics implementation produce the same density distributions? (Or, broadly what causes differences between implementations?)

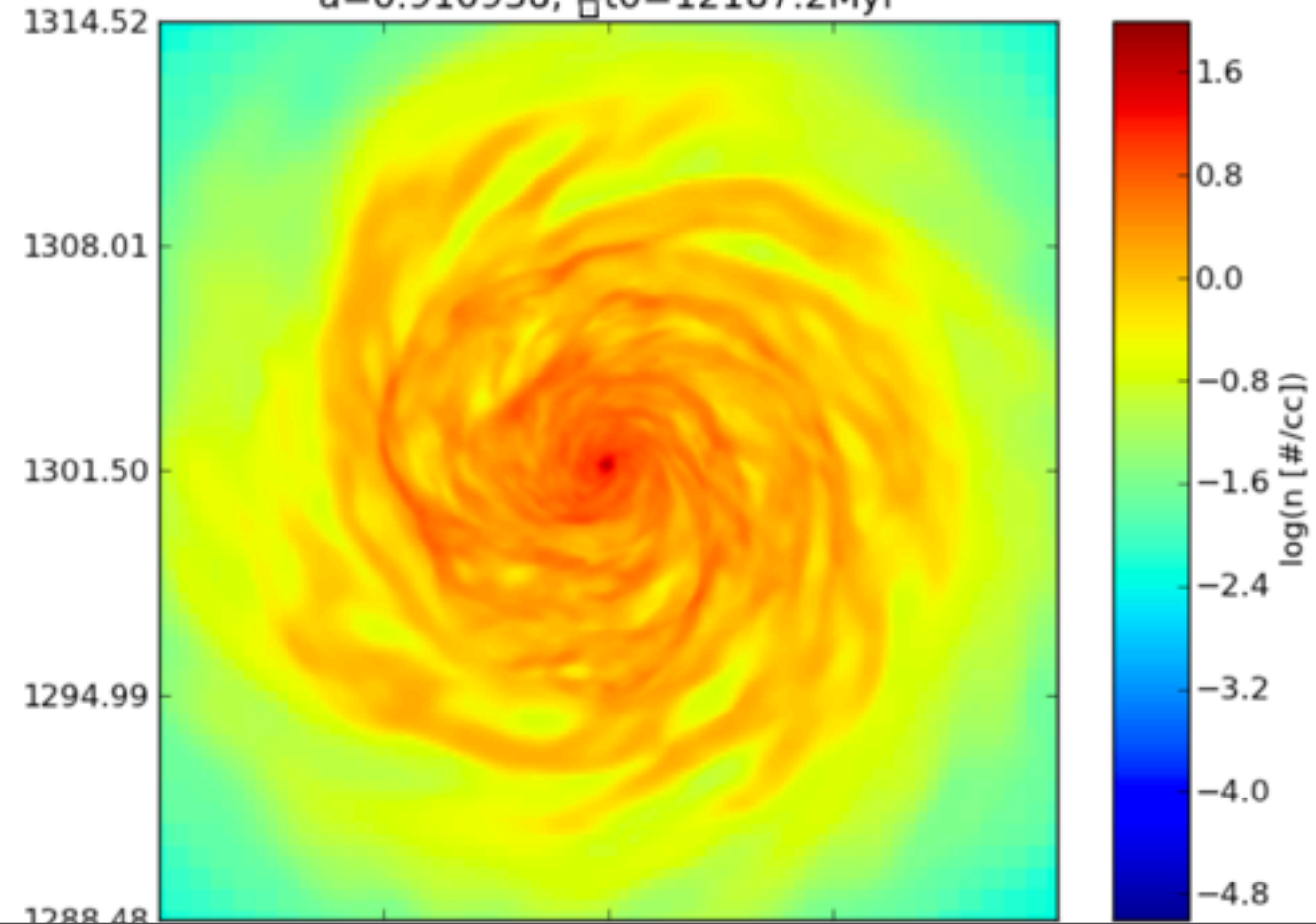
“Radiation Pressure” $30 \times L/c$
 $a=0.911138$, $t-t_0=12188\text{Myr}$



Weak thermal feedback
 $a=0.910883$, $t-t_0=12186\text{Myr}$



Extra non-thermal pressure
 $a=0.910938$, $t-t_0=12187.2\text{Myr}$



Simulation plans

- *Target:*
 - $\sim 100\text{pc}$, $m_{\text{DM}} = 10^5 M_{\text{sun}}$, quiescent $10^{12} M_{\text{sun}}$ and $10^{11} M_{\text{sun}}$.
 - Runs with two subgrid prescriptions.
 - A community AGORA subgrid model for star formation and feedback.
 - A favorite subgrid model.
 - Comparison with the isolated disk WG results.
 - Possible low redshift short re-runs to understand parameters in cosmology.
- *Next step:*
 - 600pc “proof of concept” cooling and star formation runs with common physics and favorite subgrid model.